Static & Dynamic Interconnection Networks (RGPV Notes)

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1. Introduction: Interconnection Network connects processors, memory and I/O devices. Types: Static and Dynamic.

2. Static Interconnection Network: Fixed communication links; connection pattern does not change during execution. Characteristics: Fixed connection, simple design, low cost, limited flexibility. Examples: Linear Array, Ring, Mesh, Hypercube.

3. Dynamic Interconnection Network: Communication paths created dynamically during execution. Characteristics: High flexibility, simultaneous communication, complex and expensive. Examples: Bus, Crossbar, Multistage.

4. Bus System: Shared communication line. Only one processor uses the bus at a time. Advantages: Simple, low cost, expandable. Disadvantages: Bus contention, low speed for many processors.

5. Crossbar Switch: Dedicated path between processor and memory. High speed, no bus contention, expensive. Switches required = $N \times N$.

6. Multistage Interconnection Network: Multiple stages of switches, cheaper than crossbar, scalable, may suffer blocking. Switches $\approx (N/2)\log2N$.

7. Comparison:
Bus: Shared, Low Cost, Low Speed.
Crossbar: Dedicated, Very High Speed, Very Expensive.
Multistage: Switched, Medium Cost, Excellent Scalability.

Memory Trick: Bus = One Road, Crossbar = Personal Road, Multistage = Many Small Roads.